EXECUTIVE ANALYTICS BRIEFING

MTA Ridership Analysis

Understanding Ridership Patterns, COVID-19 Impact, and Recovery

A walk through the four pages of the Power BI report.

Overview	COVID Impact	Seasonal Trends	Geographic Analysis
System totals	Baseline drop	Calendar cycles	Station density

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DATA SOURCE

Data Source and Pipeline

MTA Subway Hourly Ridership from NY Open Data, in three datasets (2017-2019, 2020-2024, 2025 onward).

Key figure: 224M hourly rows, about 12.2 billion subway entries from January 2017 to September 2026.

Architecture: medallion on a Microsoft Fabric Lakehouse

01

Bronze

The raw data exactly as it arrived, with source and load-time columns for lineage. A pipeline copied it from the source straight into Lakehouse tables, with no local downloads. Row counts matched the portal exactly.

02

Silver

Cleaned, typed hourly data in one table. Timestamps parsed, category labels standardized, and 224,430,041 rows reconciled against Bronze to the same total ridership.

03

Gold

A daily star schema, with fact_ridership_daily (about 12M rows at day × station × fare class), dim_date (with US holidays), dim_station (428 stations, stable keys), and dim_fare_class (12 classes).

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Data quality

- An integrity notebook runs 29 checks: unique keys, referential integrity, calendar gaps, and Silver-to-Gold reconciliation of ridership and transfers. Results are saved to a table after every run.
- Known source quirks are logged and left unchanged, such as a small number of negative transfer counts.

Semantic model and reporting

- A Direct Lake semantic model sits on the Gold tables, with relationships, a date table, hierarchies, sorting, and DAX measures. Report totals were verified against the Lakehouse.
- Slicer blanks were fixed with referential-integrity settings. Team access was set up through workspace roles and Build permission.

ETL pipeline orchestration

- Created a new notebook to check for updates in the source periodically and pull new records in our Bronze layer
- In case there's an update in the source, data is pulled, cleaned, type-checked, standardized in Silver, and then aggregated and modeled in the Gold layer, and the semantic model is refreshed to ensure an up-to-date

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ANALYTICAL FRAMEWORK

Ridership is more than a total

Timing, change over time, and location all shape the story.

Overview

How does ridership evolve over time?

COVID Impact

How did COVID-19 differ from the pre-pandemic baseline?

Seasonal Trends

How do patterns vary by month, weekday, weekend, and holiday?

Geographic Analysis

Where is station-entry activity concentrated?

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METRIC DEFINITION

Ridership means recorded station entries

Analytical Boundary: Ridership counts station entries—not unique passengers or complete journeys.

Source: MTA Subway Hourly Ridership, NY Open Data. Hourly entry records are reported at daily grain, with continuous coverage from January 2017 onward.

1

January 2017

Historical coverage begins

2

Daily Grain

Hourly records aggregated

REPORTING DIMENSIONS

Date & Time

Calendar date

Station Complex

Entry facility

Borough

Region

Transit Mode

Subway segment

Fare Class

Payment category

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REPORT PAGE 1

Overview: the big picture of ridership

ILLUSTRATIVE DASHBOARD LAYOUT · NO VALUES

Volume & change cards

Total and average daily ridership; year-over-year and year-to-date change.

Monthly trend

Average daily ridership line, by year.

Monthly change

Monthly ridership change columns.

Composition

Average daily ridership by borough and by fare group.

Page purpose

A system-level view of how much ridership is recorded and how it changes.

Business question

How much ridership is recorded, and how does it change over time?

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REPORT PAGE 2

COVID Impact: a baseline to measure against

Compare three fixed periods and how ridership relates to the pre-COVID level.

Pre-COVID	COVID-19	Recovery
2017–2019 baseline	2020–2021	2022 onward

Business Questions

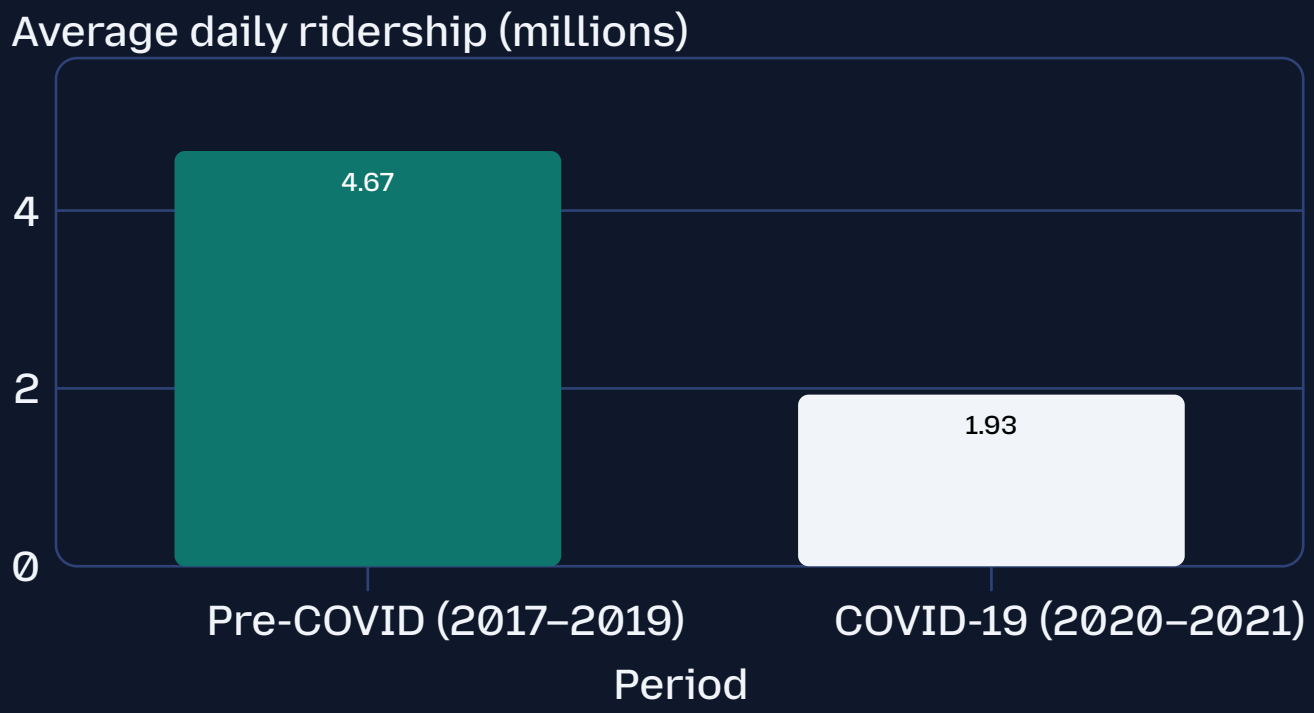
1. How much did average daily ridership decline during COVID-19?
2. How much has ridership recovered compared with the pre-COVID baseline?
3. Which boroughs have the highest and lowest recovery rates?
4. How has ridership changed over time and year over year?
5. How do year, payment method, borough, and transit mode affect the analysis?
6. What does the 30-day forecast suggest about the short-term recovery trend?

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DOCUMENTED FINDINGS

COVID-19 cut average daily ridership by 58.80%

Reported COVID Impact values; descriptive comparison, not proof of cause.



Reported average daily ridership in millions. Source: COVID Impact report view.

70.32%

Reported Recovery

System recovery as a share of the pre-COVID baseline.

BOROUGH RECOVERY RANGE

Queens: 74.18% (highest displayed)

Staten Island: 50.33% (lowest displayed)

All five boroughs remain below their documented pre-COVID baselines.

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REPORT PAGE 3

Seasonal Trends: the rhythm of ridership

Recurring calendar patterns, separate from total volume.

CONCEPTUAL CALENDAR VIEW · ILLUSTRATIVE

Calendar lens	Report view
Month	Highest/lowest month cards; index: 100 = annual daily average
Day of week	Average daily ridership
Weekday / weekend	Weekend-to-weekday ratio by month
Holiday	Holiday vs. non-holiday weekdays
Month by year	Heatmap

BUSINESS QUESTION

When does ridership tend to differ?

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REPORT PAGE 4

Geographic Analysis

Understanding Ridership Across New York City.

Business Questions

1. **Which borough has the highest total ridership?**
2. **Where are subway stations with high ridership geographically concentrated?**
3. **Which stations contribute the most to total network ridership?**

01 — System Overview

Total Ridership	Total Stations	Leading Borough	Ridership per Station
12 bn	**428**	**Manhattan**	**5M**

Interactive Filters

Borough · Station Complex · Transit Mode · Routes

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REPORT PAGE 4

Where Ridership Concentrates

Exploring Station-Level and Geographic Patterns

01 — Explore the Patterns

Station Details

A table comparing stations by borough, total ridership, and share of network.

station_complex	borough	Total Ridership	Share of Network (%)
138 St-Grand Concourse (4,5)	Bronx	8,040,161.00	0.07%
149 St-Grand Concourse (2,4,5)	Bronx	26,563,985.00	0.22%
161 St-Yankee Stadium (4,B,D)	Bronx	59,862,399.00	0.49%
167 St (4)	Bronx	20,211,743.00	0.17%
167 St (B,D)	Bronx	18,292,785.00	0.15%
170 St (4)	Bronx	17,403,947.00	0.14%
170 St (B,D)	Bronx	14,316,070.00	0.12%
174 St (2,5)	Bronx	14,124,887.00	0.12%
174-175 Sts (B,D)	Bronx	9,957,150.00	0.08%
176 St (4)	Bronx	12,461,719.00	0.10%
182-183 Sts (B,D)	Bronx	10,351,363.00	0.08%
183 St (4)	Bronx	12,988,757.00	0.11%
219 St (2,5)	Bronx	6,440,241.00	0.05%
225 St (2,5)	Bronx	7,731,204.00	0.06%
231 St (1)	Bronx	20,797,147.00	0.17%
233 St (2,5)	Bronx	9,777,297.00	0.08%
238 St (1)	Bronx	9,416,682.00	0.08%
3 Av-138 St (6)	Bronx	18,315,455.00	0.15%
3 Av-149 St (2,5)	Bronx	45,066,949.00	0.37%
Allerton Av (2,5)	Bronx	10,595,534.00	0.09%
Baychester Av (5)	Bronx	6,774,182.00	0.06%

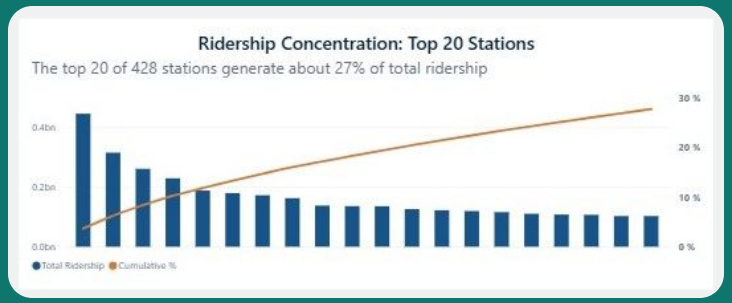
Geographic Ridership Map

A map connecting station locations with total ridership to show where subway usage is concentrated.



Ridership Concentration

A combo chart comparing ridership across the top 20 stations and showing their cumulative contribution to total network ridership.



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EXECUTIVE SYNTHESIS

Four pages, one connected story

Each page narrows the question the previous one raises.

1

PAGE 1

OVERVIEW

How much, and how it moves.

2

PAGE 2

COVID IMPACT

How it changed against a baseline.

3

PAGE 3

SEASONAL TRENDS

When it differs.

4

PAGE 4

GEOGRAPHIC ANALYSIS

Where it concentrates.

Each perspective answers a different business question.

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